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GREATER NOIDA, UTTAR PRADESH

SUMMER INTERSHIP

Data Science with Python

Beeskilled

Vikram Adhikari

Roll No.: 25SCS1003004887 | Semester III | Section: 2CSE11

B.Tech (CSE) | IILM University, Greater Noida | Batch: 2026–27

OVERVIEW

Internship Overview

ORGANIZATION	Beeskilled
DOMAIN	Data Science with Python
MODE	Online Internship
DURATION	13 July 2026 – 24 August 2026
OBJECTIVE	<i>Develop practical skills in Python-based data analysis, visualization, and introductory machine learning.</i>

Internship Objectives

- Understand Data Science using Python.
- Work with datasets using Pandas and NumPy.
- Perform data cleaning, preprocessing and statistical analysis.
- Create visualizations using Matplotlib and Seaborn.
- Perform Exploratory Data Analysis (EDA).
- Understand introductory Machine Learning and Linear Regression.
- Apply learned skills to a complete final project.

Technologies & Tools

Python	primary programming language
Pandas	data loading, cleaning and analysis
NumPy	numerical operations
Matplotlib	data visualization
Seaborn	statistical and categorical visualization
Scikit-learn	introductory machine learning and Linear Regression
Google Colab	notebook environment for the final Titanic project

Week 1 – Python for Data Science

Module: Python for Data Science Basics & Student Marks Analysis

- Introduction to Python for Data Science.
- Worked with Pandas and NumPy.
- Used the Student Marks Dataset.
- Performed basic data cleaning and statistical calculations.
- Analyzed basic data properties and results.

Week 2 – Data Visualization & EDA

Module: Data Visualization & Exploratory Data Analysis

- Worked with the COVID-19 Dataset.
- Used Matplotlib and Seaborn for visualization.
- Created trend plots and country comparisons.
- Used heatmaps and scatter plots to explore patterns and relationships.

Week 3 – Machine Learning

Module: Machine Learning & Student Performance Prediction

- Introduction to Machine Learning using Scikit-learn.
- Worked with the Student Performance Dataset.
- Performed data preparation and model training.
- Applied Linear Regression.
- Evaluated the trained model.

Week 4 – Final Project

Final Project: Titanic Dataset Analysis | Platform: Google Colab

- Loaded and inspected the Titanic Dataset.
- Handled missing values and prepared the data.
- Created AgeGroup and FamilySize features.
- Analyzed survival patterns using grouped statistics and visualizations.

Titanic Dataset – Data Preprocessing

- Dataset contained 891 passenger records.
- Missing Age values were replaced using the median.
- Cabin was removed because most values were missing.
- Missing Embarked values were filled using the mode.
- Created AgeGroup using defined age ranges.
- Created FamilySize = SibSp + Parch + 1.

Titanic Analysis – Overall Results

891

TOTAL PASSENGERS

342

SURVIVED

549

DID NOT SURVIVE

38.38%

OVERALL SURVIVAL RATE

61.62%

OVERALL DEATH RATE

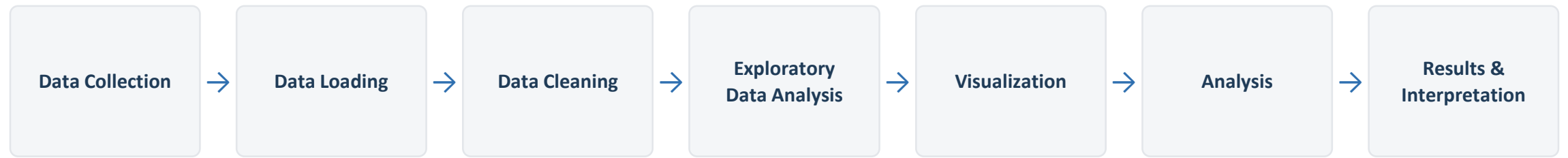
Survival was compared by gender, class, age, embarkation point and family size.

Titanic Analysis – Key Findings

BY GENDER	BY CLASS	BY AGE GROUP	BY EMBARKATION
Female	1st Class	Child	Cherbourg
74.20%	62.96%	57.97%	55.36%
Male	2nd Class	Teenager	Queenstown
18.89%	47.28%	42.86%	38.96%
	3rd Class	Young Adult	Southampton
	24.24%	35.33%	33.90%
		Adult	
		40.00%	
		Senior	
		22.73%	

Family size 4: highest observed survival rate at 72.41%.

Data Science Workflow



The internship progressively developed skills from basic data analysis to visualization, machine learning and final-project interpretation.

Skills Gained

- Python programming for Data Science
- Data cleaning and preprocessing
- Exploratory Data Analysis
- Statistical and grouped analysis
- Data visualization
- Machine Learning fundamentals
- Linear Regression using Scikit-learn
- Practical use of Google Colab
- Data interpretation and presentation

Conclusion

- The internship provided practical exposure to the basic Data Science workflow.
- Project-based learning connected Python concepts with real datasets.
- The Titanic project brought together data cleaning, feature creation, analysis and visualization.
- The internship strengthened practical analytical and problem-solving abilities.
- The actual Titanic notebook will be uploaded separately to GitHub as supporting evidence.

Certification & Official Verification



THANK YOU

Vikram Adhikari

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Data Science with Python – Beeskilled